COMPUTER NETWORKS

Week-5

Classful IP Addressing

- It divides IPv4 into fixed classes:
- **Class A**
- **Class B**
- **Class C**

IP Address Classes & Default Subnet Masks

- It divides IPv4 into fixed classes:
- **Class A** → (255.0.0.0)
- **Class B** → (255.255.0.0)
- **Class C** → (255.255.255.0)

IP Address Classes & Default Subnet Masks

- It divides IPv4 into fixed classes:
- **Class A** → /8 (255.0.0.0)
- **Class B** → /16 (255.255.0.0)
- **Class C** → /24 (255.255.255.0)

Subnet Mask

- IP address come paired with a subnet mask
- There is a default subnet mask for each IP class
  - Class A 255.0.0.0
  - Class B 255.255.0.0
  - Class C 255.255.255.0
- The subnet mask tells what part of the IP address denotes the network and which part denotes the client

Range of values In first octet	Class	Length of Network Part	Length of Host Part
1-126	A	1 octet	3 octets
128-191	B	2 octets	2 octets
192-223	C	3 octets	1 octet

CIM 2465 IP Addressing Scheme

Problems with Classful IP Addresses

Problem 1. Address wastage (if 1000 hosts are required by a company)

Class C → only 254 usable hosts (too small)

Class B → 65,534 usable hosts (too large)

Need to take a Class B network, wasting over 64,000 addresses

Problems with Classful IP Addresses

- Problem 2: No flexibility in Network Size
- Classful addressing forces organizations to choose between:
- Very small network (Class C)
- Very large network (Class B)
- Extremely large network (Class A)

- Modern networks have special needs: e.g. 800 hosts, 5,000 hosts etc.

- Solution:
- Subnetting
- Classless Interdomain Routing (CIDR)

Problems – Contd.

Problem 3: Exploding Routing Tables:

- Routing on the backbone of Internet needs to have an entry for each network address
- The size of the routing tables started to outgrow the capacity of routers

Routing table: The destination network ID, hop count to reach the network and the router ID of next hop

Problem 4: The Internet is going to outgrow the 32-bit addresses

Solution : IP Version 6

Common Problem

- Assigning an IP address to a device on your network that is already assigned to another device
- When this happens, the other computers will not know which device should get the information

Address Conflicts

- On most operating systems and devices, if there are two devices on the local network that have the same IP address → "IP Conflict" warning
- If you see this warning, that means that the device giving the warning, detected another device on the network using the same address

Solution

- Use a service called DHCP - Dynamic Host Configuration Protocol
- Provided by almost all home routers
- DHCP assigns addresses to devices and computers
- DHCP server is given the range of IP addresses you would like it to assign
- DHCP server assigns IP addresses to the various devices and keeps track of these addresses
- DHCP makes networks plug-and-play

The DHCP Process (DORA)

1. Discover

- A device (client) joins the network and broadcasts a message asking:
- “Is there any DHCP server that can give me an IP address?”
- This message is called DHCP Discover

2. Offer

- A DHCP server replies with an available IP address and configuration details
- This is called DHCP Offer
- Example offer:
- IP address: 192.168.1.25
- Subnet mask: 255.255.255.0
- Gateway: 192.168.1.1

3. Request

- The client sends a message saying: “I accept this IP address.”
- This is called DHCP Request

4. Acknowledge

- The DHCP server confirms and officially assigns the IP address
- This is called DHCP Acknowledge (ACK)

Classless IP Addressing

- To minimize the wasting of unused IP addresses in a class

→Use Sub-nets

- Introduced by Internet Engineering Task Force in 1993
- **CIDR** → Classless inter-domain routing
- To control the size of routing tables on routers
- To help slow the rapid exhaustion of IPv4 addresses

CIDR

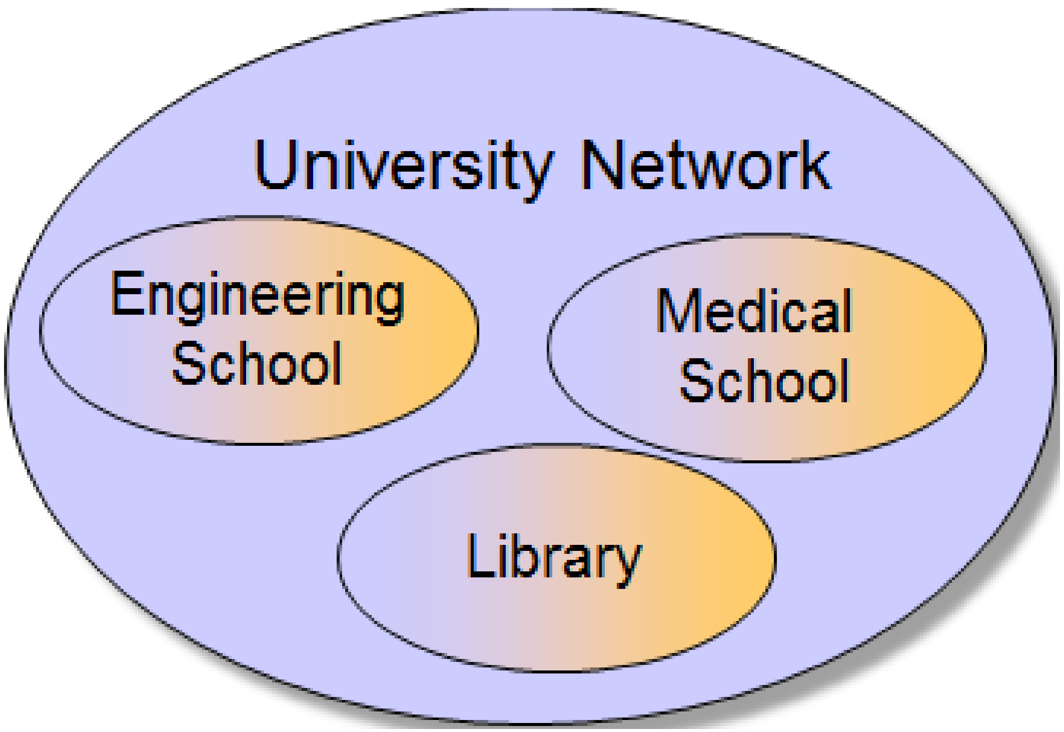
- CIDR allows:
- Custom-sized networks (/20, /22, /27, etc.)
- Efficient IP allocation
- Slower IPv4 exhaustion

Classless IP Addresses

- Consider a network that contains 9 hosts
- Only 4 bits of host suffix are needed to represent all possible host values (binary of 9 = 1001)
- However, a class C address can support 256 hosts (8-bits for host addresses: 0-255)
- Classless addressing solves this problem by allowing an ISP to assign a prefix that is 28 bits long and the suffix that is 4 bits long

Subnetting

- Organizations have multiple networks which are independently managed

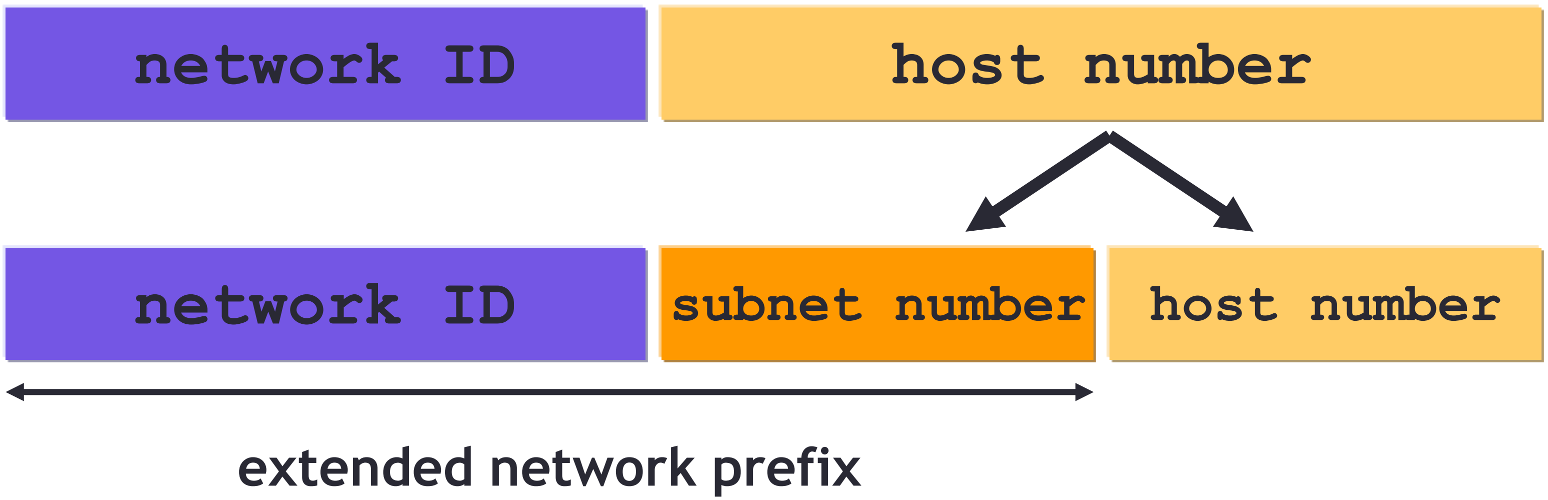


Having multiple networks

- Solution 1: Allocate an address to each network
    - Difficult to manage
    - From the outside of the organization, each network must be addressable i.e have an identifiable address
- Solution 2: Add another level of hierarchy to the IP addressing structure → Subnetting

Subnets

- Split the host number portion of an IP address into a **subnet number** and a (smaller) **host number**
- Result is a 3-layer hierarchy



Subnets

- Subnetting allows the network designer to subdivide a classful IP network into smaller groups, called subnets
- Subnets can be freely assigned within the organization
- Internally, subnets are treated as separate networks
- Subnet structure is not visible outside the organization

Subnetting

- With Subnetting, IP addresses use a 3-layer hierarchy:

  Network – Subnet - Host
- Improves efficiency of IP addresses by not consuming an entire address space for each physical network
- Reduces router complexity. Since external routers do not know about sub netting, the complexity of routing tables at external routers is reduced
- Extend Network portion, borrow from Host portion

Subnets

- A group of IP addresses that all have the same value in the first part of the address (network/block ID)
- Subnet number: A dotted-decimal number that represents a particular IP subnet

Subnetting - Rules

- Key rules
  - Devices on the same LAN must use IP addresses in the same group (**IP subnet**)
  - Devices on different LANs that are separated by at least one router must use IP addresses in different **IP subnets**
  - IP addresses must be unique inside the same internetwork

Address Masks

- Classless addressing requires an additional bit of information with each address
- This additional information specifies the exact boundary between the network prefix and the host suffix

Subnet Mask

- Depending on the network size, different values of subnet ID and host ID can be chosen
- This would prevent the outside world from being burdened by a shortage of new network addresses
- To determine the Subnetting number, a subnet *mask*—logic AND function—is used
- A subnet mask is a 32-bit number created by setting host bits to all 0s and setting network bits to all 1s. In this way, the subnet mask separates the IP address into the network and host addresses

Example

- Mask = 255.255.0.0
- 32-bit mask = 11111111 . 11111111 . 00000000 . 00000000
  (M)
- Consider a destination Address(D) : 128.10.2.3
- In binary (D) : 10000000 . 00001010 . 00000010. 00000011

- Logical AND ?

- 128 . 10 . 0.0

Example

- Mask = 255.255.0.0

Mask (M) = 11111111 . 11111111 . 00000000 . 00000000

(D) : 10000000 . 00001010 . 00000010. 00000011

(AND) : 10000000 .

Example

- Mask = 255.255.0.0

Mask (M) = 11111111 . 11111111 . 00000000 . 00000000

(D) : 10000000 . 00001010 . 00000010. 00000011

(AND) : 10000000 . 00001010 .

Example

- Mask = 255.255.0.0

Mask (M) = 11111111 . 11111111 . 00000000 . 00000000

(D) : 10000000 . 00001010 . 00000010 . 00000011

(AND) : 10000000 . 00001010 . 00000000

Example

- Mask = 255.255.0.0

Mask (M)	= 11111111 . 11111111 . 00000000 . 00000000
(D)	: 10000000 . 00001010 . 00000010 . 00000011
(AND)	: 10000000 . 00001010 . 00000000 . 00000000

Example

- Mask = 255.255.0.0

Mask (M) = 11111111 . 11111111 . 00000000 . 00000000

(D) : 10000000 . 00001010 . 00000010 . 00000011

(AND) : 10000000 . 00001010 . 00000000 . 00000000

Ntwrk Addr : 128 . 10 . 0 . 0

Example

- Mask = 255.255.0.0
- 32-bit mask = 11111111 . 11111111 . 00000000 . 00000000
  (M)
- Using a network prefix : 128.10.0.0
- Consider a destination Address(D) : 128.10.2.3
- In binary (D) : 10000000 . 00001010 . 00000010. 00000011
- Logical AND of D and M
- (M) = 11111111 . 11111111 . 00000000 . 00000000

  10000000 . 00001010 . 00000000 . 00000000
- 128 . 10 . 0.0

CIDR : Classless Inter Domain Routing

- No more classes
- Network ID part is now called Block ID
- Second part is Host ID part
- Now we can ask the user for the exact number of required host IP addresses and design our networks accordingly
- To avoid the confusion, we can now write the address notation as:

  ddd.ddd.ddd.ddd /m

- where *ddd* is the decimal value for an octet of the address, and *m* is the number of 1 bits in the mask

192.5.48.69 /26 → specifies a mask of 26 bits

CIDR

- The number of addresses requested by the customer should be the power of 2
- The block of addresses assigned must have contiguous unallocated addresses in it
- The first address of any block should be divisible by the size of the block

Practice-1 (subnets)

- Given the following address, divide this network into 2 subnets:

20.30.40.10/25

25 shows that Block ID/Network part = 25 bits

Host ID = 32-25 = 7 bits

For creating 2 subnets, we need to take 1 bit from the host ID part

Solution-1

20.30.40.10/25

Convert the last octet in binary to observe clearly

20.30.40. 00001010

To divide into 2 subnets, we need 1 bit from the host ID (blue bit)

RED is already part of the block ID/Network portion

First Subnet

20.	30.	40.	0	0	0	0	0	0	0	0	First IP Address
20.	30.	40.	0	0						1	
20.	30.	40.	0	0					1	0	
20.	30.	40.	0	0			0	0	1	1	
20.	30.	40.	0	0			0	1	0	0	
20.	30.	40.	0	0			0	1	0	1	
20.	30.	40.	0	0			0	1	1	0	
20.	30.	40.	0	0			0	1	1	1	
20.	30.	40.	0	0							
20.	30.	40.	0	0							
20.	30.	40.	0	0							
20.	30.	40.	0	0							
20.	30.	40.	0	0							
20.	30.	40.	0	0	1	1	1	1	1	1	Last IP Address
20.	30.	40.	63								

First Address: 20.30.40.0/26
Last Address: 20.30.40.63/26

2nd Subnet

20.	30.	40.	0	1	0	0	0	0	0	0	First IP Address
20.	30.	40.	0	1	0	0	0	0	0	1	
20.	30.	40.	0	1					1	0	
20.	30.	40.	0	1			0	0	1	1	
20.	30.	40.	0	1			0	1	0	0	
20.	30.	40.	0	1			0	1	0	1	
20.	30.	40.	0	1			0	1	1	0	
20.	30.	40.	0	1			0	1	1	1	
20.	30.	40.	0	1							
20.	30.	40.	0	1							
20.	30.	40.	0	1							
20.	30.	40.	0	1							
20.	30.	40.	0	1							
20.	30.	40.	0	1	1	1	1	1	1	1	Last IP Address
20.	30.	40.	127								

First Address: 20.30.40.64/26
Last Address: 20.30.40.127/26

Practice-2

- Given the following address, divide this network into 4 subnets:

20.30.40.10/25

25 shows that Block ID = 25 bits

Host ID = 32-25 = 7 bits

For 4 subnets we need to take 2 bits from the host ID part

Solution-2

20.30.40.10/25

Convert the last

20.30.40. 00001010

To divide into 4 sub nets, we need 2 bit from the host ID

Complete the solution

Subnets Details

Subnet ID	Subnet Address	Host Address Range	Broadcast Address
1	20.30.40.0	20.30.40.1 - 20.30.40.30	20.30.40.31
2	20.30.40.32	20.30.40.33 - 20.30.40.62	20.30.40.63
3	20.30.40.64	20.30.40.65 - 20.30.40.94	20.30.40.95
4	20.30.40.96	20.30.40.97 - 20.30.40.126	20.30.40.127

Example

- Given IP Address: 192.168.20.77/27
- Total IPv4 bits: 32

Example

- Given IP Address: 192.168.20.77/27
- Total IPv4 bits: 32
- Prefix Length : 27
- Host Bits : 32 – 27 = 5 host bits
- Subnet Mask ?

Example

- Given IP Address: 192.168.20.77/27
- Total IPv4 bits: 32
- Prefix Length : 27
- Host Bits : 32 – 27 = 5 host bits
- Subnet Mask :
- 11111111.11111111.11111111.11100000

- 11100000 → 224

Example

- Given IP Address: 192.168.20.77/27
- Total IPv4 bits: 32
- Prefix Length : 27
- Host Bits : 32 – 27 = 5 host bits
- Subnet Mask :
- 11111111.11111111.11111111.11100000

- 11100000 → 224
- Subnet Mask : 255.255.255.224

Example

- Given IP Address: 192.168.20.77/27
- Subnet Mask : 255.255.255.224
- Block Size : 256 – 224 = 32

Subnet Ranges (Last Octet)

- 192.168.20.0 – 31
- 192.168.20.32 – 63
- 192.168.20.64 – 95
- 192.168.20.96 – 127
- 192.168.20.128 – 159
- 192.168.20.160 – 191
- 192.168.20.192 – 223
- 192.168.20.224 – 255
- Given IP Address: 192.168.20.77/27
- 77 lies between **64 - 95**

Subnet Ranges (Last Octet)

- Given IP Address: 192.168.20.77/27
- 77 lies between **64 - 95**
- Network Address : 192.168.20.64
- Broadcast Address : 192.168.20.95
- Usable host range
- First usable: 192.168.20.65
- Last usable : 192.168.20.94
- No. of Hosts ?

Subnet Ranges (Last Octet)

- Given IP Address: 192.168.20.77/27
- 77 lies between **64 - 95**
- Network Address : 192.168.20.64
- Broadcast Address : 192.168.20.95
- Usable host range
- First usable: 192.168.20.65
- Last usable : 192.168.20.94
- No. of Hosts ?
- Host bits : 5
- 2^5 -2 = 32 – 2 = 30 hosts

Example – For practice

- Given IP: **172.16.45.200/26**

- Complete the working as done in the previous example

Ex-1

- No. of subnets needed = 14
- No. of needed usable hosts = 14
- Network Address : 192.10.10.0

- Address Class → C
- Default Subnet Mask : 255.255.255.0
- Host part : 0000 0000
- No. of bits to be borrowed : 4
- Possible Subnets → 2^4 = 16 subnets
- Custom Subnet Mask : 255.255.255.11110000

Ex-1

- Custom Subnet Mask : 255.255.255.11110000
- Custom Subnet Mask : 255.255.255.240
- Total N. of Host addresses : 2^4 = 16
- No. of usable addresses = 16 – 2 = 14

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : ?

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : C
  Default Subnet Mask : ?

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : C
  Default Subnet Mask : 255.255.255.0
- No. of bits borrowed : ?

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : C
  Default Subnet Mask : 255.255.255.0
- No. of bits borrowed : 2
- Total No. of subnets : ?

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : C
  Default Subnet Mask : 255.255.255.0
- No. of bits borrowed : 2
- Total No. of subnets : 4
- Total Number of host addresses : ?

Ex-2

- No. of required subnets : 3
- No. of required usable hosts : 45
- Given Address : 200.175.14.0
- Address Class : C
  Default Subnet Mask : 255.255.255.0
- No. of bits borrowed : 2
- Total No. of subnets : 4
- Total Number of host addresses : 2^6 = 64
- No. of usable addresses : 64 -2 = 62

Ex-3

- Given Address : 101.0.0.0
- No. of required subnets : 250
- Address Class :
- Default Subnet Mask :
- No. of bits borrowed :
- Custom Subnet mask :
- Total No. of subnets :
- Total Number of host addresses :
- No. of usable addresses :

Ex-3 - Solution

- Given Address : 101.0.0.0
- No. of required subnets : 250
- Address Class : A
- Default Subnet Mask : 255.0.0.0
- No. of bits borrowed : 8
- Custom Subnet mask : 255.255.0.0
- Total No. of subnets : 256
- Total Number of host addresses : 65,536
- No. of usable addresses : 65,534

Ex-4

- Given Address : 172.59.0.0
- No. of required subnets : 10
- Address Class :
- Default Subnet Mask :
- No. of bits borrowed :
- Custom Subnet mask :
- Total No. of subnets :
- Total Number of host addresses :
- No. of usable addresses :

Ex-4 Solution

- Given Address : 172.59.0.0
- No. of required subnets : 10
- Address Class : B
- Default Subnet Mask : 255.255.0.0
- No. of bits borrowed : 4
- Custom Subnet mask : 255.255.240.0
- Total No. of subnets : 16
- Total Number of host addresses : 4096 (2^{12})
- No. of usable addresses : 4094

Ex-5

- Given Address : 172.59.0.0
- No. of required usable hosts : 50
- Address Class :
- Default Subnet Mask :
- No. of bits borrowed :
- Custom Subnet mask :
- Total No. of subnets :
- Total Number of host addresses :
- No. of usable addresses :

Ex-5 Solution

- Given Address : 172.59.0.0
- No. of required usable hosts : 50
- Address Class : B
- Default Subnet Mask : 255.255.0.0
- No. of bits required for hosts : 6
- No. of bits borrowed : 2+8 = 10
- Custom Subnet mask : 255.255.11111111. 11000000
- Custom Subnet mask : 255.255.255.192
- Total No. of subnets : $2^{10} = 1024$
- Total Number of host addresses : $2^6 = 64$
- No. of usable addresses : 64 – 2 = 62

Book Reference

- Chapter 21
- Douglas E. Comer